UNIT 7 ELECTRICITY AND MAGNETISM

Learning objectives:

- Explain what electricity is, describe its sources (e.g., power stations, batteries, and solar panels), and understand its safe use.
- Identify the common uses of electricity in daily life, such as powering lights, appliances, and cooling/heating systems.
- Define magnets and magnetism, and differentiate between magnetic and non-magnetic materials with examples.
- Recognize the importance of electricity and magnetism in modern life and their role in powering essential devices and systems.

ELECTRICITY AND MAGNETISM

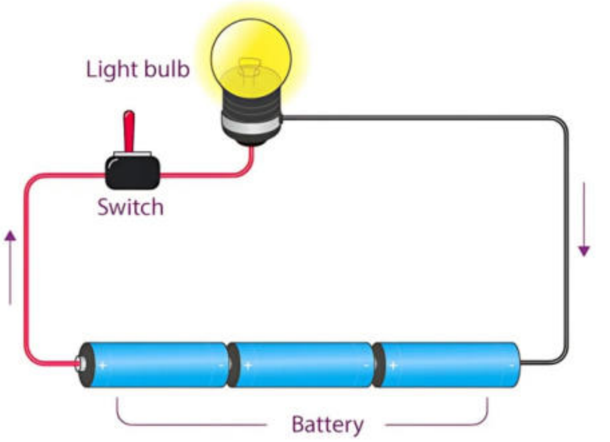
Electricity powers many devices we use daily, like lights, fans, and TVs, while magnets are special objects that attract certain metals. Understanding electricity and magnetism helps us learn how they work and how to use them safely.

WHAT IS ELECTRICITY?

Electricity is a form of energy that powers devices and machines. It flows through wires and circuits, lighting up our homes and powering appliances. We get electricity from power stations, batteries, and solar panels. While electricity is very useful, it must be handled with care as it can be dangerous.

WHAT IS ELECTRICITY?

Electricity is essential in our daily lives. Here are some common uses:



Lighting Our Homes

- Electricity powers light bulbs, tube lights, and lamps, helping us see in the dark.
- Streetlights use electricity to keep roads safe for cars and pedestrians at night.
- Electricity allows us to study, read books, and play games indoors even after sunset.

Running Fans and Air Conditioners

- In hot summers, fans circulate air, while air conditioners lower the temperature, making rooms comfortable.
- During winters, electric heaters keep homes warm and cozy.

Powering Televisions, Computers, and Other Appliances

- Electricity enables us to watch television, listen to music on the radio, and use computers for studying and entertainment.
- Refrigerators keep food fresh, washing machines clean clothes, and microwaves heat meals, all powered by electricity.
- Mobile phones and tablets need electricity to charge, allowing us to communicate, take pictures, and play games.

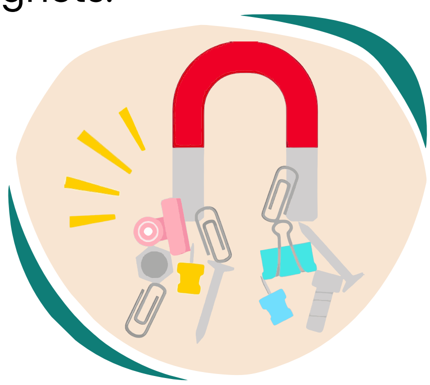
WHAT ARE MAGNETS?

Introduction to Magnets

Magnets are objects that can pull certain metals toward them. This pulling force is called magnetism. If you've ever played with magnets, you've noticed how they attract small metal objects like pins, nails, or paper clips. Magnets attract metals like iron, nickel, and cobalt.

What Do Magnets Attract?

- Magnets do not attract everything. They only pull objects made of certain metals.
- **Magnetic Materials:** Iron, steel, nickel, and cobalt are examples of materials magnets can attract.
- **Non-Magnetic Materials:** Materials like plastic, rubber, glass, and wood are not attracted by magnets.



Fill in the blanks.

1. Electricity is a form of ______________ that powers devices and machines.

2. Streetlights help keep roads ______________ at night.

3. ______________ and ______________ are devices we use to cool down in the summer.

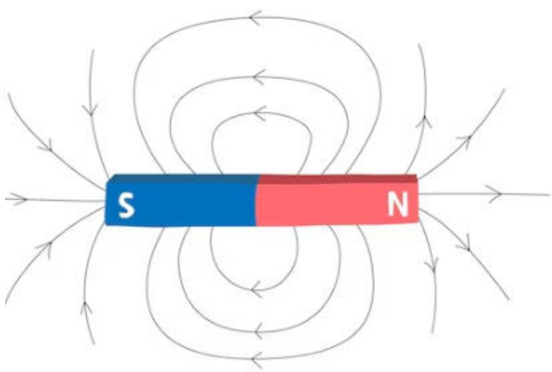
4. ______________ materials like iron and steel are attracted to magnets.

5. ______________ materials like plastic and rubber are not attracted to magnets.

6. Magnets produce a pulling force called ______________ .

Magnetic fields

BAR MAGNET



Write "T" for True or "F" for False.

1. Magnets can attract plastic objects.

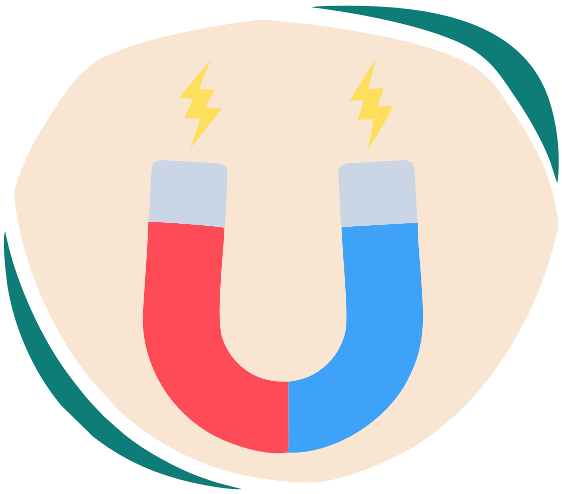
2. Electricity flows through circuits and powers appliances.

3. Fans and air conditioners are used to keep homes warm.

4. Iron and nickel are examples of magnetic materials.

5. Electricity must be used carefully because it is dangerous.

6. Non-magnetic materials include rubber and wood.



Choose the correct answer.

1. What powers the lights in our homes?

- Magnets
- Electricity
- Water
- Solar panels

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2. Which of the following is a magnetic material?

- Plastic
- Iron
- Rubber
- Wood

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3. What is the pulling force produced by magnets called?

- Electricity
- Magnetism
- Gravity
- Friction

4.What do refrigerators use to keep food fresh?

- Magnets
- Water
- Electricity
- Wind

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5. Which device is used to cool homes during the summer?

- Air conditioner
- Refrigerator
- Heater
- Microwave

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6. What is a non-magnetic material?

- Steel
- Nickel
- Wood
- Iron

Critical Thinking

What would happen if there were no electricity in the world?

How would it affect our daily lives?

• Encourage students to discuss how electricity is essential for lighting, cooling, communication, and other activities.

Answer the following questions.

Q1: What is electricity, and why must it be used carefully?

Q2: Name three ways we use electricity in our homes.

Q3: What are magnets, and what do they attract?

Q4: What is the difference between magnetic and non-magnetic materials?

Q5: How does electricity help us in hot summers and cold winters?

Q6: What are some devices that use electricity, and how do they help us?

Create a Simple Circuit

ACTIVITY

- **Objective**: Understand how electricity flows in a circuit.
- **Materials Needed**: A battery, a small bulb, wires, and a switch.
- **Instructions**:

1. Connect the bulb to the battery using wires.
2. Add a switch to control the flow of electricity.
3. Turn the switch on and off to see the bulb light up.

- **Outcome**: Students will learn how electricity flows through a circuit to power a device.